

Experiments :- Study of LAN Topology  
Network device and Address configuration

Objective :-

- 1) To study the LAN topology used in college campus.
- 2) To identify networking device & understand their function
- 3) To analyze wired & wireless communication method used in the network.
- 4) To measure speed of network for wired & wireless connections.
- 5) To obtain MAC, IP Address & subnet mask.

→ Problem statement - 1

- Draw & identify LAN topology of college campus & clearly label all components such as router, switch & etc.

Solve :- The college campus network observed this study  
Consists of

- \* Layer - 3 Core switches in the server room
- \* Managed by PoE & Switches
- \* Wireless Access point (APs)
- \* Ethernet port in classroom & Laboratories

## → Network topology

Hybrid Hierarchical topology which combines :

- 1) Tree topology
- 2) Extended star topology
- 3) Ring topology (server room)

## → The network hierarchical layer

- ① Core layer
- ② Distribution layer
- ③ Access layer

### ① Core layer

\* Core layer contains server room and server room have firewall, Router, 2 - L3 + switches, Patch Rack server inverter, etc

### ② Distribution layer

\* Distribution layer contains 4 - switches

\* The 4 switches are :-

- ① Main Build → R7 (Room no 213, 2nd floor)
- ② Main Build → R1 (Room no - 301, 3rd floor)
- ③ Main Build → R6 (Room no - 113, 1st floor)
- ④ Main Build → R2 (Room no - 201, 2nd floor)

This 4 switches contains

- ① 18 Port Gigabit Easy smart switch with 16 port

TL - SG1218MPE

- ② SG3428XMP → 24 port, Gigabit 2 4-port  
10 GE SFP, 12+ managed switches with 24 port port

→ Now let's discuss these 2 above components :-

### ① TP - Link TL - SG1218MPE

- \* Likely used for : CCTV Camera wifi access point etc
- \* 18 port Gigabit "Easy smart" Switch
- \* 16 port support port

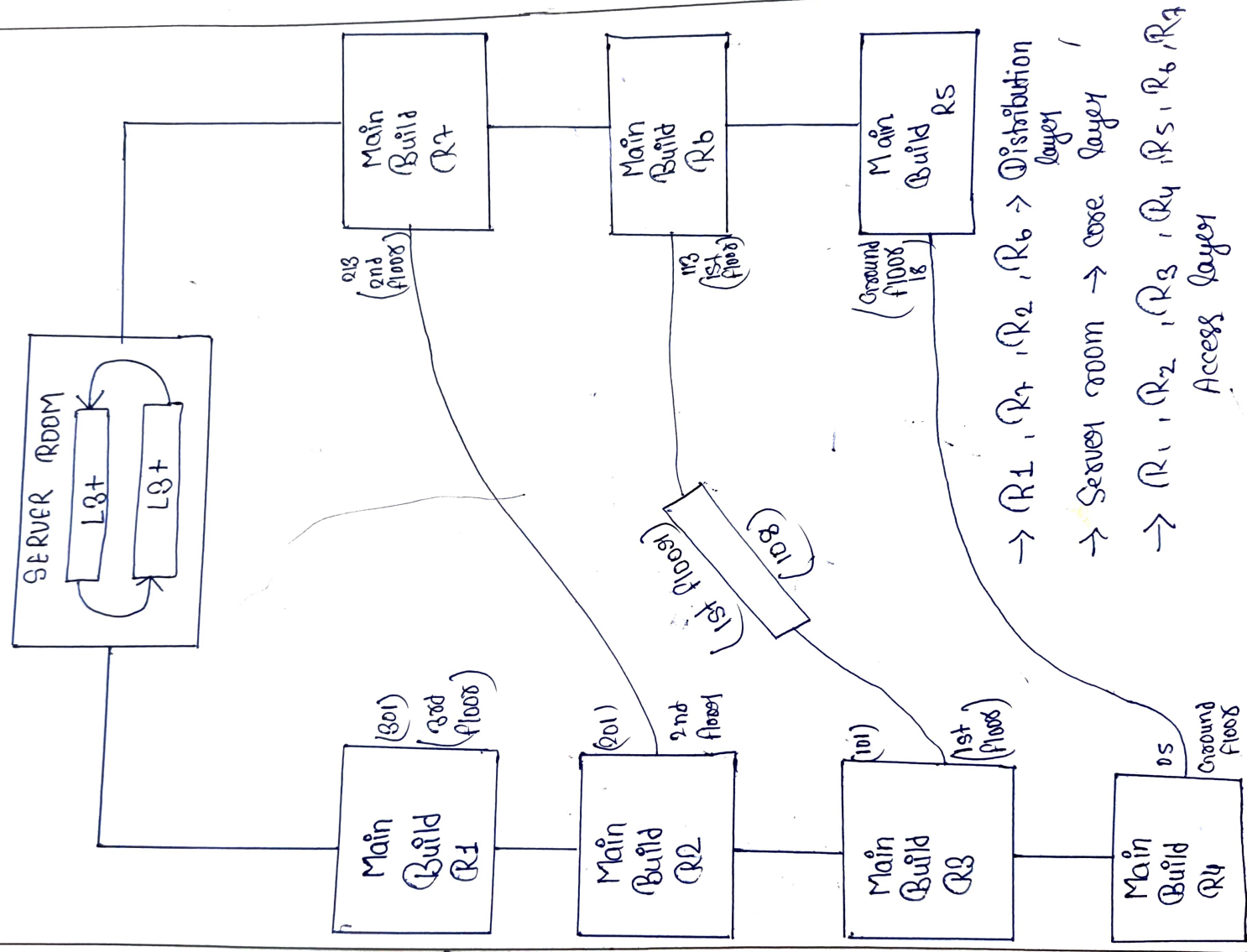
### ② TP - Link Omada SG3428XMP

- \* Highest - end managed layer 2 + switches
- \* 24 port port, uplink, 10G SFP
- \* usually acts as a main distribution for each floor.

### → Access layer

- \* Access layer - Contain 3 - switches i.e.
  - 1) Main Build R1 - ROOM no - 301, 3rd floor]
  - 2) Main Build R4 - [ROOM no - 05, Ground floor]
  - 3) Main Build R5 - [ROOM no - 18 Ground floor]

Topology





(7)

→ Difference b/w wired & wireless speed?

### Wired Network

- \* Faster Communication
- \* Stable Connection
- \* Low latency
- \* Dedicated physical medium

### Wireless Network

- \* Slower communication
- \* Affected by distance / interface
- \* Higher latency
- \* Shared radio medium

→ Ethernet provide direct physical communication using network cables resulting in higher speed & stable performance.

→ Wireless Connection uses radio waves which are affected by interference, distance, obstacles & multiple connected users.

Command Prompt

## Windows IP Configuration

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::992c:6cd2:c438:a76c%8  
IPv4 Address. . . . . : 192.168.56.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection\* 9:

Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection\* 10:

Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::3e82:c0b0:3123:d815%5  
IPv4 Address. . . . . : 10.118.32.243  
Subnet Mask . . . . . : 255.255.255.0

WIRELESS CONNECTION

↳ using "ipconfig"

Command Prompt

Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . : fe80::992c:6cd2:c438:a76c%8  
IPv4 Address. . . . . : 192.168.56.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection\* 9:

Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection\* 10:

Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . : fe80::3e82:c0b0:3123:d815%5  
IPv4 Address. . . . . : 10.118.32.243  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 10.118.32.2

WIRED CONNECTION

C:\Users\ADMIN>

→ using IPConfig

\* Every Switches act as access layer which help to provi network connectivity to all classroom Lab faculty Room , etc.

\* Access layer directly connect

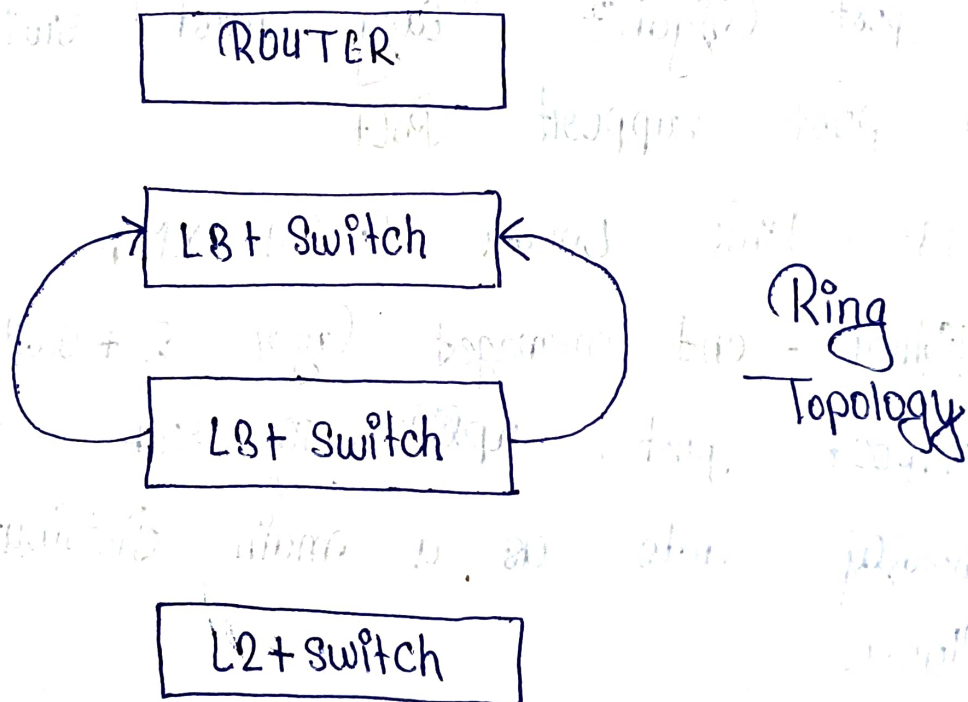
① Ethernet port

② Wireless Ap

③ Lab System

④ outdoor wireless (AP (EX - 018))

→ Server Room Architecture / topology





Q) Role of subnet mask in network?

→ A subnet mask identifies the network portion & host portion of an IP address. It helps device determine whether another device belongs to the same network or different network.

\* It is important for:-

- ① Network segment
- ② Efficient routing
- ③ Communication b/w devices

## → OBSERVATION

During the study of the college network infrastructure.

- ① A centralized server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-Link PoE switches were used in network cabinets

→ Advantage of existing network ?

- \* Scalable architecture
- \* Centralized management.
- \* Support PoE device.
- \* Easy expansion of floor & classroom.
- \* Support both wireless & wired
- \* Redundant link for failover

### Conclusion

The observed campus LAN follow a hybrid hierarchical topology consist of tree, extended star, Ring. The network uses managed PoE switch, wireless access Point Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.

Microsoft Windows [Version 10.0.22621.4317]  
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C:\Users\ADMIN>getmac

| Physical Address  | Transport Name                                       |
|-------------------|------------------------------------------------------|
| B0-0C-D1-2C-3A-DF | Media disconnected                                   |
| E6-16-CA-83-CF-DB | \Device\Tcpip_{27635621-3133-4B8B-9255-E7C4BB9E0EA7} |
| 0A-00-27-00-00-08 | \Device\Tcpip_{3F28BB86-23C7-4F82-8497-6D44C1776D07} |

C:\Users\ADMIN>

→ MAC ADDRES

→ using "getmac"



Processes

Performance

App history

Startup apps

Users

Details

Services

Performance

CPU

25% 1.94 GHz

Memory

4.9/7.8 GB (63%)

Disk 0 (C:)

SSD

18%

Ethernet

Ethernet 2

S: 0 R: 0 Kbps

Wi-Fi

Wi-Fi

S: 0 R: 72.0 Kbps

GPU 0

Intel(R) HD Graphics 6...

1%



Run new task



Wi-Fi

Throughput

Intel(R) Dual Band Wireless-AC 8265

500 Kbps

450 Kbps

Send

0 Kbps

Receive

72.0 Kbps

Adapter name: Wi-Fi

SSID: GECK-Campus-WiFi

Connection type: 802.11n

IPv4 address: 10.85.23.183

IPv6 address: fe80::502a:354f:4ef7:85a%5

Signal strength:



Settings